

1) Find 10 different model providers and their models released in last 12 months

| S. No | Model Provider  | Model                     | Release Date | Knowledge Cutoff | API Cost (Input / Output per 1M Tokens) | Context Window    | Type         | When to Use This Model                                                               |
|-------|-----------------|---------------------------|--------------|------------------|-----------------------------------------|-------------------|--------------|--------------------------------------------------------------------------------------|
| 1     | OpenAI          | GPT-5 .5                  | Apr 2026     | Dec 2025         | \$5 / \$30                              | 1M                | Proprietary  | Complex reasoning, coding, research, enterprise applications <a href="#">OpenAI</a>  |
| 2     | Anthropic       | Claude Sonnet 4.5         | Feb 2026     | Jan 2025         | ~\$3 / \$15                             | 1M                | Proprietary  | Long documents, coding, enterprise assistants <a href="#">FixAEQ</a>                 |
| 3     | Google DeepMind | Gemini 3.1 Pro            | Feb 2026     | Jan 2025         | Varies by Vertex AI                     | 2M                | Proprietary  | Multimodal AI, search, reasoning, video understanding <a href="#">FixAEQ</a>         |
| 4     | xAI             | Grok 4.3                  | Apr 2026     | Dec 2025         | ~\$3 / \$15                             | 256K              | Proprietary  | Real-time knowledge, coding, X platform integration <a href="#">FixAEQ</a>           |
| 5     | Meta            | Llama 4 (Scout /Maverick) | Apr 2026     | Not Published    | Free (Open Weights)                     | Up to 10M (Scout) | Open Weights | Self-hosting, fine-tuning, research, private deployments <a href="#">Tom's Guide</a> |
| 6     | Alibaba         | Qwen 3.8-Max              | Aug 2026     | Not Published    | Provider-specific                       | 1M                | Open Weights | Multilingual AI, enterprise chatbots, coding <a href="#">Reuters</a>                 |
| 7     | Mistral AI      | Magistral Medium          | Jun 2026     | Not Published    | Provider-specific                       | 128K              | Proprietary  | Fast reasoning, enterprise applications <a href="#">FixAEQ</a>                       |
| 8     | DeepSeek        | DeepSeek V4 Flash         | Aug 2026     | Not Published    | \$0.14 / \$0.28                         | 1M                | Open Weights | Lowest-cost inference, coding, large-scale deployments <a href="#">Reuters</a>       |
| 9     | Moonshot AI     | Kimi K2.5                 | Jul 2026     | Not Published    | Provider-specific                       | Up to 1M          | Proprietary  | Long-context reasoning, document analysis <a href="#">Reuters</a>                    |
| 10    | Z.ai (Zhipu AI) | GLM-4 .6                  | Jul 2026     | Not Published    | Provider-specific                       | 128K+             | Open Weights | Chinese & multilingual applications, enterprise AI <a href="#">FixAEQ</a>            |

2) Read 2–3 articles about how a transformer works?

Goal to understand:

- (a) Self-attention mechanism
- (b) Vector embedding
- (c) Feed forward Neural Network

Transformer -> *A transformer is a type of artificial intelligence model that learns to understand and generate human-like text by analyzing patterns in large amounts of text data.*